R-UNIVERSALITY OF A UNITARY ALGEBRA OVER A QUATERNION DIVISION ALGEBRA IN TERMS OF THE REDUCED NORM AND THE EVEN CLIFFORD ALGEBRA

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Allowed standard prerequisites (editorial).

- The Tits algebra of the natural representation over a quaternion division algebra with the reversion anti-involution. Used inside the counted proof at source lines 741-742, attributed by the author to page 379 of the cited book KnusEtAl-BookOfInvols. Identifies the first Tits algebra with the quaternion division algebra itself.
- The even Clifford algebra of a Hermitian form and the behaviour of the remaining Tits algebras according to the parity of the size. Used in the proof where the author compares the two remaining classes and uses that the centre of the spin group has exponent two in the even case (source lines 762-764). Produces the two alternatives of the statement, according to whether the size is even or odd.

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Source reference. Original lemma 2.1: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 416–416. Navigation pointer only: the counted proof invokes the author's equivalence for R-universality; that lemma has its own card in this lane and its content is declared as a named standard prerequisite.

Source reference. Original definition 4: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 432–432. Navigation pointer only: the labelled item of that equivalence used by the counted proof.

Source reference. Original definition 3: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 428–428. Navigation pointer only: the labelled item of that equivalence used by the counted proof.

Source reference. Original lemma 2.3: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 471–471. Navigation pointer only: the author's lemma on the minimal inner extension; it has its own card in this lane and its content is declared as a named standard prerequisite.

Source reference. Original proposition 5.1: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 674–674. Navigation pointer only: the counted proof refers to that proposition's setup; it is counted by this lane's card P1-fl02-so-even-clifford-split and its text is not reproduced here.

Lemma 0.1. *Let $\mathfrak{g} = \mathfrak{su}_k(B; D, \tau_r)$, where*

- *D is a quaternion division algebra over $\mathbb{Q}$,*
- *τ_r is the reversion anti-involution, and*
- *B is a τ_r-Hermitian matrix in $\mathrm{GL}_k(D)$.*

Then $\mathfrak{g}$ is $\mathbb{R}$-universal if and only if D does not split over $\mathbb{R}$, and either

1. *k is even and the reduced norm of B (calculated in the algebra $\mathrm{Mat}_{k\times k}(D)$) is a square in $\mathbb{Q}$, or*
2. *k is odd and $C_D^0(B, \tau_r)$ is split.*

Proposition 0.2 ([11, Corollary 3.5, § 4.2, and Lemma 7.4])**.** *Let L be the center of the algebra $D_{\mathfrak{g},F}(\lambda) := \mathrm{End}_{\mathfrak{g}}({}^F\rho_\lambda)$ (which, by Schur's lemma, is a division algebra). Then:*

1. *$\mathrm{Gal}(\overline{F}/L) = \{\sigma \in \mathrm{Gal}(\overline{F}/F) \mid \sigma(\lambda) = \lambda\}$ (for the $*$-action of $\mathrm{Gal}(\overline{F}/F)$), and*
2. *$\beta_{\mathfrak{g},L}(\lambda) = [D_{\mathfrak{g},F}(\lambda)]$, after identifying $H^2(L; \boldsymbol{\mu})$ with the Brauer group of L.*

Proof. As in the proof of Proposition 5.1, let λ, λ_+, λ_- be dominant weights that represent the three nonzero elements of Z^*, with λ being the highest weight of the natural representation of $\mathfrak{g}$ on $(D \otimes_{\mathbb{Q}} \mathbb{C})^k$.

($\Rightarrow$) We have $D_{\mathfrak{g},\mathbb{Q}}(\lambda) = D$ [4, p. 379], so D does not split over $\mathbb{R}$ (see Lemma 2.1(3)). (Note that this implies $\mathfrak{g} \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathfrak{so}_k(\mathbb{H})$.)

Suppose k is even, so $\mathfrak{g}\otimes_{\mathbb{Q}}\mathbb{R}$ is inner. Then $\mathfrak{g}$ must also be inner (see Lemma 2.3). Since k is even, this means that the reduced norm of B is a square in $\mathbb{Q}$.

Suppose k is odd. Then $\mathfrak{g} \otimes_{\mathbb{Q}} \mathbb{R}$ is outer, so the weights λ_+ and λ_- are not fixed by the $*$-action. Therefore $D_{\mathfrak{g},\mathbb{R}}(\lambda_\pm) = \mathbb{C}$ (see Proposition 0.2). Since $D_{\mathfrak{g},\mathbb{Q}}(\lambda_\pm)$ is Brauer equivalent to $C_D^0(B, \tau_r)$ [4, p. 379], we conclude that $\mathfrak{g}$ is $\mathbb{R}$-universal if and only if this algebra is split (see Lemma 2.1(4)).

($\Leftarrow$) The proof for odd k is reversible, so let us assume k is even. Then $D_{\mathfrak{g},\mathbb{R}}(\lambda_\pm) = \mathbb{H}$ is nontrivial. (For example, $D_{\mathfrak{g},\mathbb{R}}(\lambda_\pm)$ can be calculated by using [11, § 5.5].) Therefore $D_{\mathfrak{g},\mathbb{Q}}(\lambda_\pm)$ does not split over $\mathbb{R}$. Since $D_{\mathfrak{g},\mathbb{Q}}(\lambda_\pm)$ is a quaternion algebra (because the center of $\mathbf{Spin}_{2k}$ has exponent 2 when k is even), we conclude that $D_{\mathfrak{g},\mathbb{Q}}(\lambda_\pm) \otimes_{\mathbb{Q}} \mathbb{R} \cong D_{\mathfrak{g},\mathbb{R}}(\lambda_\pm)$, as desired. □

References

[1] Demazure M., Grothendieck A., Schémas en groups. III. Structure des schémas en groupes reductifs, Séminaire de géométrie algébrique du Bois Marie, 1962/1964, available at `http://library.msri.org/books/sga/sga/pdf/sga3-3.pdf`.

[2] Harder G., Bericht über neuere Resultate der Galoiskohomologie halbeinfacher Gruppen, *Jber. Deutsch. Math.-Verein.* **70** (1968), 182–216, available at `http://www.digizeitschriften.de/en/dms/img/?PPN=PPN37721857X_0070&DMDID=dmdlog17`.

[3] Kneser M., Lectures on Galois cohomology of classical groups, *Tata Institute of Fundamental Research Lectures on Mathematics*, Vol. 47, Tata Institute of Fundamental Research, Bombay, 1969, available at `http://www.math.tifr.res.in/~publ/ln/tifr47.pdf`.

[4] Knus M.A., Merkurjev A., Rost M., Tignol J.P., The book of involutions, *American Mathematical Society Colloquium Publications*, Vol. 44, Amer. Math. Soc., Providence, RI, 1998.

[5] Lam T.Y., Introduction to quadratic forms over fields, *Graduate Studies in Mathematics*, Vol. 67, Amer. Math. Soc., Providence, RI, 2005.

[6] Morris D., Real representations of semisimple Lie algebras have $\mathbb{Q}$-forms, in Algebraic Groups and Arithmetic, Tata Institute of Fundamental Research, Mumbai, 2004, 469–490.

[7] Platonov V., Rapinchuk A., Algebraic groups and number theory, *Pure and Applied Mathematics*, Vol. 139, Academic Press, Inc., Boston, MA, 1994.

[8] Prasad G., Rapinchuk A., On the existence of isotropic forms of semi-simple algebraic groups over number fields with prescribed local behavior, *Adv. Math.* **207** (2006), 646–660.

[9] Raghunathan M.S., Arithmetic lattices in semisimple groups, *Proc. Indian Acad. Sci. Math. Sci.* **91** (1982), 133–138.

[10] Tits J., Classification of algebraic semisimple groups, in Algebraic Groups and Discontinuous Subgroups (Proc. Sympos. Pure Math., Boulder, Colo., 1965), Amer. Math. Soc., Providence, R.I., 1966, 33–62.

[11] Tits J., Représentations linéaires irréductibles d'un groupe réductif sur un corps quelconque, *J. Reine Angew. Math.* **247** (1971), 196–220.

[12] Tits J., Strongly inner anisotropic forms of simple algebraic groups, *J. Algebra* **131** (1990), 648–677.